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Extracted Text

the eye at a: fo that none of the rays that fall above _h_ on below i, can be reflected to the eye at o; and the dif. stance between _6 and i is equal to half the length of the object _AB._

Hence it appears, that if a man fees like whole image in a plane looking glats, the part of the glats that reflects his image melt be jull half as long and half as broad as himself, let him fland at any distance from it whatever; and that this image mull appear just as far behind the glafs as he is before it. Thus, the man AB (No. 30.) viewing himself in the plane mirrored CD, which is jull half as long as himself, fees his whole image as at _EF_, behind the glafs exactly equal to his own fzz. For, a ray AC, proceeding from his eye at A, and falling perpendicularly upon the surface of the glafs at G, is re-flected back to his eye, in the frame line CaI : and the eye of this image will appear at B, in the frame line pro-duced to E, beyond the glafs. And a ray BD, wrong from his foot, and falling obliquely on the glafs at D, will be reflected as obliquely on the other side of the perpendicular AD), in the direction D,K, and the foot of this image will appear at Ft, in the direction of the re-fled ray AD, produced to F, where is is cut by the right line ACE, drawn parallel to the right line ACE. Jull the fans as if the glafs were taken away, and a real man flood at F, equal in five to the man (handing as B: for to his eyes at A, the eye of the other man at E would be few in the direction of the line ACE; and the foot of the man as F would be seen by the eye A, in the direrection of the line ADF.

Of the MAGIC LANTERN.

ABCD (No. 31.) is a tin lantern, with a tube nklm fixed in the field of it. This tube conflicts of two joints, one of which dips into the others and by drawing this joint out, or pulling it in, the tube may be made longer or shorter. At kl in the end of the moveable joint of the tube, a convex lens is fixed, and an object painted with transparent colors upon a piece of thin glafs is pla- edat de homewhere in the immovable joint of the tube; fo that as the tube is lengthened or fluorescent, the lens will be either at a greater or a left distance from this transferen object. In the field of the lantern three is a very convex lens bbc, which ferries to call a very frong light from the candle within the lantern upon the object de. New when the rays, which fringe through the objet de, diverge from the several points as Å, a, sc. in the object, and fall upon the lens kl, they will be made to converge to as many points fj, g &e. on the other ride of the lens, and will point an inverted picture of the

Vox. III. No ğ7.

object at fg upon a white wall, a fiecet or a fireen of white paper, provided the object is farther from the less than its principal focus. To make this picture appear, dificent and brighty, it must have no other light fall up- on it but what comes through the lens kl ; and for this reason the whole apparatus is to be placed in a dark, room EFGH. The lens A mull be very convex, fo that the object de may be very near to it, and yet not be near- orthan its principal focus; for by this means, as the object is near to the lens, the picture fg will be at a great distance from it , and consequently the picture will be much bigger than the object. Since the picture is inverted in respect of the object, in order to make the picture appear with the right end upwards, it is neceflary that the object de should be placed with the wrong end upwards.

Of the DIFFERENT REFRANGIBILITY of LIGHT.

We have hitherto toppofed that a particle of light, as it comes from the fun, is the leaf particular iron which light can be separated. But we mull now correct this top-polition, by thawing, that a particle of light, as it comes from the fun, is, properly speaking, a bundle of rays, which may be farated from one another. Therefore, for the future, by a ray of light we mull be understood to mean, not that collection of particles which we have hitherto called by this name, but the leaf particles in- to which light can be farated.

Rays of light are laid to be differently refrangible, when an the form or equal angles of incidence forms are more turned out of the way than others.

Rays are laid to be differently refasible, if firms are more safely reflected than others.

Light is called homogeneous, when all the rays are e- quality refrangible: i is treated heterogeneous, when time rays are other refrangible than others.

The colours of homogeneous light are called primary or female colours ; and thief of heterogeneous light are called secondary or mixed.

The rays of the fun are not all equally refrangible: and thief rays, which have a different degree of re-fragibility, here likewise a different colors.

If a beam of light SF, (No. 32.) that comes from the fun, paths into a dark room through F a round hole in a window fluter EG ; this beam proceeding straight forwards, and falling upon a paper as Y, would there make a round picture of the fun. This picture would be a confufed one indeed, if the hole is a large one and there is no lens in the hole. However, as this round f spotlight a a picture of the fun, we shall hereafter, notwithflanking its condition, call it by this name. Now it a glafs prifm ABC is placed between the hole in the window-fluiter and the paper at Y, the rays of this term; by the refraction which they differ in the prifm, will be bent from their fruit courte, and instead of going to io as to fall upon the paper at Y, they will be turned upwards, and the picture of the fun produced by them will fall up- on a paper MN that is placed above Y. If all the rays **OYI (447)** **OUS** 1. Aries, or ram and fleep, the horos of which are flu- ped like a half moon, and compreffed. This animal is perhaps the molt gentle and notifintive, and at the fame time the molt timid and fliquid, of all quadrupeds. In a flock of fleep, there is always a leader, whole flesh the cell amplified and blindly follow. If the men over a precipice, the whole flock follow his example, and cannot be reiftained though evident defiruation be the confe- engence. The finalist buffs, if uncommon, makes them run precipitantly against each other, without knowing the cause of their danger, or belogable to extricate them, folves. In favor, they remain fixed in the fate fots and continue disbursing in this situation, under they are forced to move by the flhepherd or his dog. But through daftally in itself, though altro devoid of feminine and mental qualities, this animal is of the molt extective uitality to man. It infacts on both the and climbing, fer- files the advantages we derive from its milk, its flow, and its rollow.

Love, the lifeliftentment in all animals, is the on- ly pathion which infiples the ram with any degree of vi- vacity; when under the influence of this pallon, he bo- come prettien, more significant and sometimes even attacks the flhepherh himself. But the ew, so fit- male, though in the fame litution, does not appear to be in the last affected; it the items to have no other difficult than that of simply admitting the embraces of the ram, eating her food, and recognizing her offspring; the lamb, on the other hand, is endowed with the fame inflint of diliguishing its means from the self of the dock.

This animal, fo fo and lo simple in its nature and disposition, is likewise extremely weak and feeble in its condition: They can endere but little fingers when ever they ran, their hearts pute, and their wind falls there, they are equally incapable of clearing her or cold, fawor or rain: They are fuject to many difades, molt of which the consensus: They bring forth their young with pain and difficulty, and require more care and attention than any other scientific animal.

The rains capable of generations at the aged 18 months; and the cow carbe impregnated when a year old. One ram is sufficient for go or upo even. He ought to be flrog, well proportioned; his head should be thick and floreg, his front large, his eyes block, his mole fits, his neck thick, his body long and tall, has articles muffy, and his tail long. White is the belt colour for a ram. The ew whose wool is molt plentiful, bulhy, long, fot, and white, the most proper for breeders. Especially when at the fame time they are of a large size, have a thick neck, and move rimbly.

cancer, in azobby. See Leo.

Ewes are generally in fields from the mof of the obber to the beginning of April. However, they may be impeg- nated as any time, it they be fed with illuminating non- rilmen, as fat water, and bread made of hem-feed. They go with your about, few months, and generally bring forth, but one at a time, though

frequently two. In warm climates, they may bring forth twice in a year, but in Britain, France, and molt parts of Europe, only enough. They goes milk plentyfully for tenx or eight minutes: Their milk affords mouthments for children broad country people; good checefe is made of it, especially women mixed with cow-milk. They live from 10 to 12 thyvers, and are capable of bringing forth an long as they tailites.

When the mile limbs are not intended to be keys for propagation, but farmed for food, they might to be calibrated at the age of five or fix months. This operation is performed two ways: in the one, an inclion is made and the facilities taken on; in the other, a liature is tied by around the forearm, above the facilities, which foot deftroys the veffels which nourish them. After ca- affirates, they are called wearless.

The new, ewes, and weather, when one year old, lofs the two foreseeh of the under jaw; fix months afterwards, they lofe the two foreseeh next to thele; and at the age of three years, the teeth are all replaced. The age of a ram may likewise be delivered by their home, which always appears the left year, and frequently saboon as they are brought forth. Their horse uniformly ac- quite an additional ring every year as longer the creature lives. The ews commonly have an horses, but a kind of long protubrances in place of them: However, fome of them have two, and some four hors.

In Spain, and the Southern passa of Europe, the flocks are kept in thudes or fiables during the night; but in Britain, where there is now no danger from wolves, they are allowed to remain without; both adult and day; which make the animals more healthy, and their dish a more whole food. Dry and mountainous grounds, where thyme and fleep's fefea grafs abound, are the built for the patturing fleep.

2. The Guinemands, or Guinea fleep, has pendulous ears, lax hairy dew lips, and a prominence on the hind part of the head. The weol is short, like that of a goat. It is a native of Guinea.

3. The fireplaces, or Cretan fleep, has fruit ceri- nated horses, twilled in a firal manner. It is a native of Mount folia.

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the eye at *o*: ſo that none of the rays that fall above *h*, or below *i*, can be reflected to the eye at *o*; and the diſtance between *h* and *i* is equal to half the length of the object *AB*.

Hence it appears, that if a man ſees his whole image in a plane looking-glaſs, the part of the glaſs that reflects his image muſt be juſt half as long and half as broad as himſelf, let him ſtand at any diſtance from it whatever; and that his image muſt appear juſt as far behind the glaſs as he is before it. Thus, the man *AB* (No. 30.) viewing himſelf in the plane mirrour *CD*, which is juſt half as long as himſelf, ſees his whole image as at *EF*, behind the glaſs, exactly equal to his own ſize. For, a ray *AC*, proceeding from his eye at *A*, and falling perpendicularly upon the ſurface of the glaſs at *C*, is reflected back to his eye, in the ſame line *CA*; and the eye of his image will appear at *E*, in the ſame line produced to *E*, beyond the glaſs. And a ray *BD*, flowing from his foot, and falling obliquely on the glaſs at *D*, will be reflected as obliquely on the other ſide of the perpendicular *abD*, in the direction *DA*; and the foot of his image will appear at *F*, in the direction of the reflected ray *AD*, produced to *F*, where it is cut by the right line *BGF*, drawn parallel to the right line *ACE*. Juſt the ſame as if the glaſs were taken away, and a real man ſtood at *F*, equal in ſize to the man ſtanding at *B*: for to his eye at *A*, the eye of the other man at *E* would be ſeen in the direction of the line *ACE*; and the foot of the man at *F* would be ſeen by the eye *A*, in the direction of the line *ADF*.

If the glaſs be brought nearer the man *AB*, as ſuppoſe to *cb*, he will ſee his image as at *CDG*: for the reflected ray *CA* (being perpendicular to the glaſs) will ſhew the eye of the image as at *C*; and the incident ray *Bb*, being reflected in the line *bA*, will ſhew the foot of his image as at *G*; the angle of reflection *abA* being always equal to the angle of incidence *Bba*: and ſo of all the intermediate rays from *A* to *B*. Hence, if the man *AB* advances towards the glaſs *CD*, his image will approach towards it; and if he recedes from the glaſs, his image will alſo recede from it.

Of the Magic Lantern.

ABCD (No. 31.) is a tin lantern, with a tube *nklm* fixed in the ſide of it. This tube conſiſts of two joints, one of which ſlips into the other: and by drawing this joint out, or puſhing it in, the tube may be made longer or ſhorter. At *kl* in the end of the moveable joint of the tube a convex lens is fixed, and an object painted with tranſparent colours upon a piece of thin glaſs is placed at *de* ſomewhere in the immoveable joint of the tube; ſo that as the tube is lengthened or ſhortened, the lens will be either at a greater or a leſs diſtance from this tranſparent object. In the ſide of the lantern there is a very convex lens *bbc*, which ſerves to caſt a very ſtrong light from the candle within the lantern upon the object *de*. Now when the rays, which ſhine through the object *de*, diverge from the ſeveral points as *d*, *e*, &c. in the object, and fall upon the lens *kl*, they will be made to converge to as many points *f*, *g*, &c. on the other ſide of the lens, and will paint an inverted picture of the object at *fg* upon a white wall, a ſheet or a ſcreen of white paper, provided the object is farther from the lens than its principal focus. To make this picture appear diſtinct and bright, it muſt have no other light fall upon it but what comes through the lens *kl*; and for this reaſon the whole apparatus is to be placed in a dark room EFGH. The lens *kl* muſt be very convex, ſo that the object *de* may be very near to it, and yet not be nearer than its principal focus: for by this means, as the object is near to the lens, the picture *fg* will be at a great diſtance from it, and conſequently the picture will be much bigger than the object. Since the picture is inverted in reſpect of the object, in order to make the picture appear with the right end upwards, it is neceſſary that the object *de* ſhould be placed with the wrong end upwards.

Of the Different Refrangibility *of* Light.

We have hitherto ſuppoſed that a particle of light, as it comes from the ſun, is the leaſt particle into which light can be ſeparated. But we muſt now correct this ſuppoſition, by ſhewing, that a particle of light, as it comes from the ſun, is, properly ſpeaking, a bundle of rays, which may be ſeparated from one another. Therefore, for the future, by a ray of light we muſt be underſtood to mean, not that collection of particles which we have hitherto called by this name, but the leaſt particles into which light can be ſeparated.

Rays of light are ſaid to be differently refrangible, when at the ſame or equal angles of incidence ſome are more turned out of the way than others.

Rays are ſaid to be differently reflexible, if ſome are more eaſily reflected than others.

Light is called homogeneous, when all the rays are equally refrangible: it is called heterogeneous, when ſome rays are more refrangible than others.

The colours of homogeneous light are called primary or ſimple colours; and thoſe of heterogeneous light are called ſecondary or mixed.

The rays of the ſun are not all equally refrangible: and thoſe rays, which have a different degree of refrangibility, have likewiſe a different colour.

If a beam of light SF, (No. 32.) that comes from the ſun, paſſes into a dark room through F a round hole in a window ſhutter EG; this beam proceeding ſtraight forwards, and falling upon a paper at Y, would there make a round picture of the ſun. This picture would be a confuſed one indeed, if the hole is a large one and there is no lens in the hole. However, as this round ſpot of light is a picture of the ſun, we ſhall hereafter, notwithſtanding its confuſion, call it by this name. Now if a glaſs priſm ABC is placed between the hole in the window-ſhutter and the paper at Y, the rays of this beam, by the refraction which they ſuffer in the priſm, will be bent from their ſtrait courſe, and inſtead of going on ſo as to fall upon the paper at Y, they will be turned upwards, and the picture of the ſun produced by them will fall upon a paper MN that is placed above Y. If all the rays

Figure 1: *
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1. Aries, or ram and ſheep, the horns of which are ſhaped like a half moon, and compreſſed. This animal is perhaps the moſt gentle and inoffenſive, and at the ſame time the moſt timid and ſtupid, of all quadrupeds. In a flock of ſheep, there is always a leader, whoſe ſteps the reſt implicitly and blindly follow. If he runs over a precipice, the whole flock follow his example, and cannot be reſtrained though evident deſtruction be the conſequence. The ſmalleſt noiſe, if uncommon, makes them run precipitantly againſt each other, without knowing the cauſe of their danger, or being able to extricate themſelves. In ſnow, they remain fixed in the ſame ſpot; and continue obſtinately in that ſituation, unleſs they are forced to move by the ſhepherd or his dog. But though daſtardly in itſelf, though almoſt devoid of ſentiment and mental qualities, this animal is of the moſt extenſive utility to man. It affords us both food and cloathing, beſides the advantages we derive from its milk, its ſkin, and its tallow.

Love, the livelieſt ſentiment in all animals, is the only paſſion which inſpires the ram with any degree of vivacity; when under the influence of this paſſion, he becomes petulant, runs againſt his neighbours, and ſometimes even attacks the ſhepherd himſelf. But the ewe, or female, though in the ſame ſituation, does not appear to be in the leaſt affected; ſhe ſeems to have no other inſtinct than that of ſimply admitting the embraces of the ram, eating her food, and recogniſing her offſpring; the lamb, on the other hand, is endowed with the ſame inſtinct of diſtinguiſhing its mother from the reſt of the flock.

This animal, ſo ſoft and ſo ſimple in its nature and diſpoſition, is likewiſe extremely weak and feeble in its conſtitution: They can endure but little fatigue; whenever they run, their hearts palpitate, and their wind fails them; they are equally incapable of bearing heat or cold, ſnow or rain: They are ſubject to many diſeaſes, moſt of which are contagious: They bring forth their young with pain and difficulty, and require more care and attention than any other domeſtic animal.

The ram is capable of generation at the age of 18 months; and the ewe can be impregnated when a year old. One ram is ſufficient for 30 or 40 ewes. He ought to be ſtrong, well proportioned; his head ſhould be thick and ſtrong, his front large, his eyes black, his noſe flat, his neck thick, his body long and tall, his teſticles maſſy, and his tail long. White is the beſt colour for a ram. The ewes whoſe wool is moſt plentiful, buſhy, long, ſoft, and white, are moſt proper for breeders, eſpecially when at the ſame time they are of a large ſize, have a thick neck, and move nimbly.

Ewes are generally in ſeaſon from the end of October to the beginning of April: However, they may be impregnated at any time, if they be fed with ſtimulating nourithment, as ſalt water, and bread made of hemp-ſeed. They go with young about five months, and generally bring forth but one at a time, though frequently two: In warm climates, they may bring forth twice in a year; but in Britain, France, and moſt parts of Europe, only once. They give milk plentifully for ſeven or eight months: Their milk affords nouriſhment for children and country people; good cheeſe is made of it, eſpecially when mixed with cow-milk. They live from 10 to 12 years, and are capable of bringing forth as long as they live.

When the male lambs are not intended to be kept for propagation, but fattened for food, they ought to be caſtrated at the age of five or ſix months. This operation is performed two ways: in the one, an inciſion is made and the teſticles taken out; in the other, a ligature is tied tight round the ſcrotum, above the teſticles, which ſoon deſtroys the veſſels which nouriſh them. After caſtration, they are called weathers.

The ram, ewe, and weather, when one year old, loſe the two foreteeth of the under jaw; ſix months afterwards, they loſe the two foreteeth next to theſe; and at the age of three years, the teeth are all replaced. The age of a ram may likewiſe be diſcovered by their horns, which always appear the firſt year, and frequently as ſoon as they are brought forth. Theſe horns uniformly acquire an additional ring every year as long as the creature lives. The ewes commonly have no horns, but a kind of long protuberances in place of them: However, ſome of them have two, and ſome four horns.

In Spain, and the ſouthern parts of Europe, the flocks are kept in ſhades or ſtables during the night: but in Britain, where there is now no danger from wolves, they are allowed to remain without, both night and day; which makes the animals more healthy, and their fleſh a more whole food. Dry and mountainous grounds, where thyme and ſheep's feſcue graſs abound, are the beſt for the paſturing ſheep.

2. The Guineenſis, or Guinea ſheep, has pendulous ears, lax hairy dew laps, and a prominence on the hind part of the head. The wool is ſhort, like that of a goat. It is a native of Guinea.

3. The ſtrepſiceros, or Cretan ſheep, has ſtrait carinated horns, twiſted in a ſpiral manner. It is a native of Mount Iola.

OULNEY, a market-town of Buckinghamſhire, ſituated nine miles ſouth-eaſt of Northampton.

OULX, a town of Italy, in the province of Piedmont, ſituated in E long. 6° 30', N. lat. 45°.

OUNCE, a little weight, the ſixteenth part of a pound avoirdupois, and the twelfth part of a pound troy.

OUNCE, in zoology. See LEO.

OUNDLE, a market-town of Northamptonſhire, ſituated on the river Nen, twenty-two miles north-eaſt of Northampton.

OVOLO, or OVUM, in architecture, a round moulding, whoſe profile, or ſweep, in the ionic and compoſite capitals, is uſually a quadrant of a circle: whence it is alſo commonly called the quarter-round.

It is uſually cut with the repreſentation of eggs and anchors or arrows heads placed alternately.

OUSE, a river, which, riſing in the north of Yorkſhire, runs ſouth-eaſt by York; and, continuing that courſe, falls into the Trent.

OUSE, is alſo a river which riſes on the confines of Oxfordſhire and Buckinghamſhire; and running north eaſt through Buckinghamſhire, Bedfordſhire, Huntingdonſhire, Cambridgeſhire, and Norfolk, paſſes by Buckingham, Bedford, Huntington, and Ely, diſcharging itſelf into the bay of the German ſea at Lynn.

OUSTIACH, or OSTIACH *Country*, is a part of Aſiatic Ruſſia, extending along the river Irtis to its confluence with the river Oby, and from thence northward along the banks of the Oby and Jeniſa, into the gulph of the Mangaſea, or the frozen ocean; and extending alſo along

Figure 2: *
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